Improve the law of learning

HW3 - Pham Minh Trí - 2353235

A1. One perceptron update

Weights $w = (w_0, w_1, w_2) = (0.1, -0.3, 0.4)$. Input $x = (1, 1, -1)$ (bias $x_0 = 1$), target $t = -1$, $\eta = 0.2$

$$z = w^T x = 0.1(1) + (-0.3)(1) + 0.4(-1) = -0.6$$

Since $z < 0$, the prediction is $\hat{y} = -1$, which equal target $t = -1$, so the example classified correctly.

The update is $\Delta w = \eta (t - \hat{y}) x = 0.2 \times 0 \times x = (0, 0, 0)$.

No update needed, the weights stay $(0.1, -0.3, 0.4)$

A2. Activation functions as design choices

z	-3	-0.5	0	2
$\text{ReLU} = \max(0, z)$	0	0	0	2
$\sigma(z) = \frac{1}{1+e^{-z}}$	0.0474	0.3775	0.5	0.8808

- Range: $\sigma \in (0, 1)$; $\text{ReLU} \in [0, +\infty)$
- Saturation: $\sigma' \to 0$ for large $|z|$ (max 0.25) => vanishing gradient. ReLU does not saturate for $z > 0$ (derivative 1).
- Zero gradient: ReLU has derivative exactly 0 for $z < 0$, so a unit that is negative for all inputs stops learning (dying ReLU). Sigmoid's gradient may be tiny but never exactly zero.
- Situation where tanh is preferable: tanh is zero-centered output in $(-1, 1)$. Prefered in a hidden layer so the next layer's inputs are not all positive => weight gradients do not share one sign (less zigzag)

A3. Trace a tiny backpropagation step

Given: $x = 1$, $w = 0.6$, $v = -0.7$, zero biases, $t = 0$,

$$L = \frac{1}{2}(t - o)^2$$

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Forward pass:

$z_h = wx = 0.6 \quad h = \sigma(0.6) \approx 0.6457$

$z_0 = vh = -0.7(0.6457) \approx -0.4520$

$o = \sigma(-0.4520) \approx 0.3889$

$L = \frac{1}{2}(0 - 0.3889)^2 \approx 0.0756$

Output error signal:

$\delta_0 = \frac{\partial L}{\partial z_0} = (o - t)o(1-o) = 0.3889 \times 0.3889 \times 0.611$

≈ 0.0924

Update direction for w:

$\delta_h = \delta_0 vh(1-h) = 0.0924 \times (-0.7) \times (0.6457 \times 0.3543)$

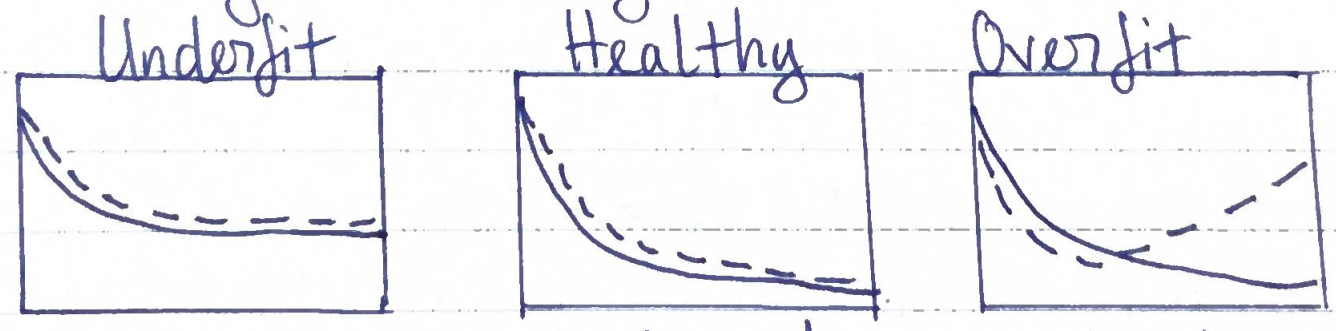
≈ -0.0148

$\frac{\partial L}{\partial w} = \delta_h x \approx -0.0148 < 0$

Gradient descent: $w \leftarrow w - \eta(-0.0148)$, so w increase.

Reason: $o = 0.3889$ is < target $0 \Rightarrow z_0 = vh$ need to lower. Since $h > 0 \Rightarrow$ make v more negative. Since $v < 0$ $\Rightarrow h \uparrow \rightarrow z_0 \downarrow$ and $h \uparrow$ when $w \uparrow$.

A4. Diagnose training curves



Solid = training loss, dashed = validation loss

- Underfit: both high, close → more units/layers, train longer.
- Healthy: both low, small gap → keep; early stopping
- Overfit: train ↓, val ↑ → early stopping, L2/dropout, smaller net, more data

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AS. Assignment bridge: controlled capacity experiment.
Setup: Fixed: standardised features, official subject-disjoint split, Adam optimiser, $\alpha = 10^{-4}$, max_iter = 100, seed 42, metrics accuracy and macro-F1.
Changed only: hidden width

Model	Train acc	Test acc	Test F1	Time(s)	Params
A	0.9978	0.9491	0.9496	7.39	36,358
B	0.9989	0.9440	0.9439	8.98	72,710

Params: A: $561 \times 64 + 64 + 64 \times 6 + 6 = 36358$
B: $561 \times 128 + 128 + 128 \times 6 + 6 = 72710$
Prediction: B higher train, test similar or slightly lower, runtime 1.2 - 1.4x
Result:
- Train: 99.78% -> 99.89% as predicted
- Test acc: 94.91% -> 94.40%, drop 0.51 points
- Runtime: x1.215 as predicted

=> No mismatch
- Plausible reason: 64 units already suffice for 561 engineered features; extra width may fit training subject habits. However 15 windows with one seed is within noise: A and B are not clearly different. More seed are needed.

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